

PhD Candidate – Computer Science / Computer Architecture

in @alexandrejamet

- Major subjects: mathematics, physics, and chemistry.

ALEXANDRE VALENTIN JAMET

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PUBLICATIONS

- A. V. Jamet ^{id}, G. Vavouliotis ^{id}, M. Torrents ^{id}, D. Chasapis ^{id} and M. Casas ^{id}, "Enhancing Instruction Prefetching via Cache and TLB Management". To appear in the 53rd IEEE International Symposium on Computer Architecture (ISCA), June 2026.
- A. V. Jamet ^{id}, G. Vavouliotis ^{id}, L. Alvarez ^{id}, D. A. Jiménez ^{id} and M. Casas ^{id}, "Practically Tackling Memory Bottlenecks of Graph-Processing Workloads". Published in the 38th IEEE International Parallel & Distributed Processing Symposium (IPDPS), May 2024.
 - 📄 Paper: 10.1109/IPDPS57955.2024.00096
 - 📄 Slides: 10.5281/zenodo.11398316.
- A. V. Jamet ^{id}, G. Vavouliotis ^{id}, L. Alvarez ^{id}, D. A. Jiménez ^{id} and M. Casas ^{id}, "A Two Level Neural Approach Combining Off-Chip Prediction with Adaptive Prefetch Filtering.". Published in the 2024 IEEE International Symposium on High-Performance Computer Architecture (HPCA), Edinburgh, United Kingdom, 2024, pp. 528-542, doi:
 - 📄 Paper: 10.1109/HPCA57654.2024.00046
 - 📄 Slides: 10.5281/zenodo.10789477.
 - ✂ Poster: 10.5281/zenodo.10797183.
 - 📄 Artifact: 10.5281/zenodo.10113336.
- A. V. Jamet ^{id}, L. Alvarez ^{id}, D. A. Jiménez ^{id} and M. Casas ^{id}, "Characterizing the Impact of Last-Level Cache Replacement Policies on Big-Data Workloads". Published in the 2020 IEEE International Symposium on Workload Characterization (IISWC), Beijing, China, 2020, pp. 134-144, doi:
 - 📄 Paper: 10.1109/IISWC50251.2020.00022.
 - ▶ Presentation: Watch the presentation on YouTube.
 - 📄 Slides: 10.5281/zenodo.10563544.

WORKSHOPS

- A. V. Jamet ^{id}, L. Alvarez ^{id} and M. Casas ^{id}, "Characterizing the impact of last-level cache replacement policies on big-data workloads". 10th Barcelona Supercomputing Center Doctoral Symposium, May 2023. doi:
 - 📄 Extended Abstract: 10.48550/arXiv.2305.06696.

PATENTS

- European Patent Application EP 4575807 A1 – "Perceptron-Based Off-Chip Predictor" Inventors: A. V. Jamet, G. Vavouliotis, M. Casas. Filed Dec 2023, published Jun 2025.

ACHIEVEMENTS

- 🏆 **Sport**
Participated in the 2018 Finswimming World Championships.
- 🏠 **Non-profit Organization**
Raised lottery funds for finswimming equipment and international competition trips.

TECHNICAL SKILLS

Performance Engineering C/C++
Computer Architecture Assembly
Shell Python VHDL OpenMP
Jupyter Notebooks GDB/LLDB
Clang Stack

LANGUAGES

French
English
Spanish



EDUCATION

PhD Candidate

Barcelona Supercomputer Center /
Universitat Politècnica de Catalunya

📅 2018 – 2024 📍 Barcelona, Spain

- Worked on developing predictors leveraging AI-inspired methods.
- Published research papers in top-tier international conferences on Computer Architecture: IPDPS, HPCA, and IISWC.

Graduate School of Engineering

TELECOM Nancy, France

📅 2015 – 2018 📍 Nancy, France

- Major in Computer Sciences, minor in Embedded Software.
- Focus on Computer Architecture.

Preparatory Classes to Graduate Schools

LGT Baimbridge, Guadeloupe

📅 2013 – 2015 📍 Les Abymes, Guadeloupe

- Major subjects: mathematics, physics, and chemistry.